*American Journal of Obstetrics & Gynecology, Research Letter. Body text below is 4 sections (Objective, Study Design, Results, Conclusion), 5 references, 1 figure, Supplemental Methods and Table moved to online supplementary material.*

# Timing of GLP-1 Receptor Agonist Fills Relative to Conception Among Commercially Insured Women

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**Manuscript word count (text only):** 499. **Study type:** cohort study.

**Keywords:** glucagon-like peptide-1 receptor agonist; semaglutide; tirzepatide; preconception; periconception exposure; pregnancy; discontinuation; unintended pregnancy; administrative claims; MarketScan

**Tweetable statement (optional, under 210 characters):** GLP-1 fills among commercially insured women with live births generally declined before conception and fell sharply after. Counseling on pregnancy intention belongs at initiation, not at diagnosis.

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**Statement of Authorship:** to be signed by all authors at submission.

**Data sharing:** Merative MarketScan data are licensed from the vendor and cannot be redistributed by the authors; the analytic code will be made available on request.

**Objective.** Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) are not recommended in pregnancy, and labeling advises discontinuing semaglutide at least 2 months before conception.(1) GLP-1 utilization during the year leading up to delivery has grown over time.(2) But timing of GLP-1 RA cessation relative to conception has not been described. We measure GLP-1 utilization and discontinuation rates in event time surrounding conception for commercially insured women with a live birth.

**Study Design.** Using Merative MarketScan Commercial Claims and Encounters data from 2017–2025, we constructed a study population of women ages 15–49 with an inpatient live-birth 2019–2025 using established claims algorithms (Supplemental Methods). We dated conception using Z3A gestational-age diagnosis codes linked to delivery admission (Supplemental Methods).[1] We restricted the sample to people with continuous enrollment with a pharmacy benefit from 12 months before to 3 months after conception (N=649,193). Each annual delivery cohort consisted of a balanced longitudinal panel with monthly observations anchored to the calendar month of conception. We defined a binary variable indicating any pharmacy claim for a GLP-1 RA during each month. We computed the fraction of the sample with a GLP-1 fill each month relative to conception separately by delivery year. In the sub-sample of patients with GLP-1 use during the 365 days before conception, we also computed days from last fill to conception. This study was determined exempt by the Indiana University Institutional Review Board.

**Results.** Pre-conception GLP-1 utilization grew from $<.05\%$ in 2019 to $>1\%$ in 2025 (Figure). Utilization rose across the preconception year in the 2023–2025 cohorts, while earlier cohorts (2019–2021) remained flat. GLP-1 utilization fell substantially after conception. In the 2025 cohort, GLP-1 utilization fell from 1.06% in the conception month to 0.39% 1 month after conception and 0.10% 3 months after conception. In the 2019 cohort, GLP-1 utilization fell from 0.04% in the conception month to 0.02% 1 month after conception and $<0.01\%$ 3 months after conception.

Among 6,067 deliveries with any preconception-year fill, 35.90% last filled within 30 days of conception and 33.81% had at least 1 fill in the conception month or the following 2 months (Supplemental Table).

**Conclusion.** Most of the decline in GLP-1 RA utilization followed conception, placing cessation at pregnancy recognition. Fewer than 1 in 4 women used contraception at GLP-1 RA treatment initiation, suggesting some pregnancies may be unplanned.(5) Periconception exposure has not been associated with major congenital malformations among women with type 2 diabetes,(3) but one large cohort study reported that prepregnancy or early-pregnancy discontinuation was associated with greater gestational weight gain, preterm delivery, gestational diabetes, and hypertensive disorders.(4) The 2-month interval is specific to semaglutide; labeling for other GLP-1 RAs instead addresses contraceptive effectiveness at initiation and dose escalation (Supplemental Methods). Contraception and pregnancy-intention counseling should occur at GLP-1 RA initiation and dose escalation, not only at pregnancy identification. Limitations include conditioning on live birth, dispensing rather than ingestion, exclusion of  5% of deliveries without a valid gestational-age code, partial ascertainment in 2025, and uncaptured compounded and cash-pay products.

[1] Results were robust to alternative conception-date approaches (Supplemental Figure)

## References (5, the AJOG Research Letter limit)


1. Wegovy (semaglutide) injection, for subcutaneous use. Prescribing information. Novo Nordisk Inc; x. Accessed x. x

2. Lessard C, Cary C, In A, et al. Prescribing trends in glucagon-like peptide-1 medications among pregnant and postpartum persons. *Obstet Gynecol.* 2026;147(3):290–292.

3. Cesta CE, Rotem R, Bateman BT, et al. Safety of GLP-1 receptor agonists and other second-line antidiabetics in early pregnancy. *JAMA Intern Med.* 2024;184(2):144–152.

4. Maya J, Pant D, Fu Y, et al. Gestational weight gain and pregnancy outcomes after GLP-1 receptor agonist discontinuation. *JAMA.* 2025;334(24):2186–2196.

5. Thapaliya K, Sweeting A, Black KI, Poprzeczny A, Mazza D, Grzeskowiak LE. Incidence of GLP-1 receptor agonist use by women of reproductive age attending general practices in Australia, 2011–2022: a retrospective open cohort study. *Med J Aust.* 2025;223(7):365–371.

## Supplemental Methods

**Pregnancy episode identification.** Inpatient live-birth delivery episodes were identified using the claims algorithm of Ailes et al.(S1) restricted to admissions with a liveborn-specific Z37 diagnosis among women ages 15–49. The inpatient admission date was used as the delivery-date proxy in all analyses.

**Conception dating.** We use each person's Z3A gestational-age diagnosis code linked to the delivery admission to estimate the date of conception. We define conception date by subtracting (gestational age in weeks $-$ 2) $\times$ 7days from the delivery admission date. Delivery episodes without a Z3A code (about 5% of raw candidates) are excluded. As a robustness check, we compared this gestational-age approach with a fixed interval approach validated by Margulis et al.(S2). They found that subtracting 273 days from the date of birth best estimated the last menstrual period for deliveries without preterm codes. Assuming conception occurs approximately 14 days after the last menstrual period, we used a fixed 259-day interval (admission date minus 259 days, applied uniformly regardless of gestational age). We compared the two dating methods both among all episodes with valid Z3A evidence and among the subset with a full-term (39–40 week) gestational age. Results were qualitatively similar across dating method and cohort restriction (Supplemental Figure).

**Labeled discontinuation intervals.** Semaglutide labeling instructs discontinuation at least 2 months before a planned pregnancy because of the long half-life of semaglutide.(1) Tirzepatide labeling specifies no preconception interval; it instructs discontinuation when pregnancy is recognized and advises patients using oral hormonal contraceptives to switch to a non-oral method or add a barrier method for 4 weeks after initiation and after each dose escalation.(S3) The 60-day concordance measure in the Supplemental Table is therefore reported for the full cohort and, separately, restricted to episodes whose last preconception fill was semaglutide.

## Supplemental references


S1. Ailes EC, Zhu W, Clark EA, et al. Identification of pregnancies and their outcomes in healthcare claims data, 2008–2019: an algorithm. *PLoS One.* 2023;18(4):e0284893.

S2. Margulis AV, Setoguchi S, Mittleman MA, Glynn RJ, Dormuth CR, Hernandez-Diaz S. Algorithms to estimate the beginning of pregnancy in administrative databases. *Pharmacoepidemiol Drug Saf.* 2013;22(1):16–24.

S3. Zepbound (tirzepatide) injection, for subcutaneous use. Prescribing information. Eli Lilly and Company; x. Accessed x. x

## Figure. GLP-1 Receptor Agonist Fills Relative to Conception, by Delivery Year, Among Commercially Insured Women With a Live Birth

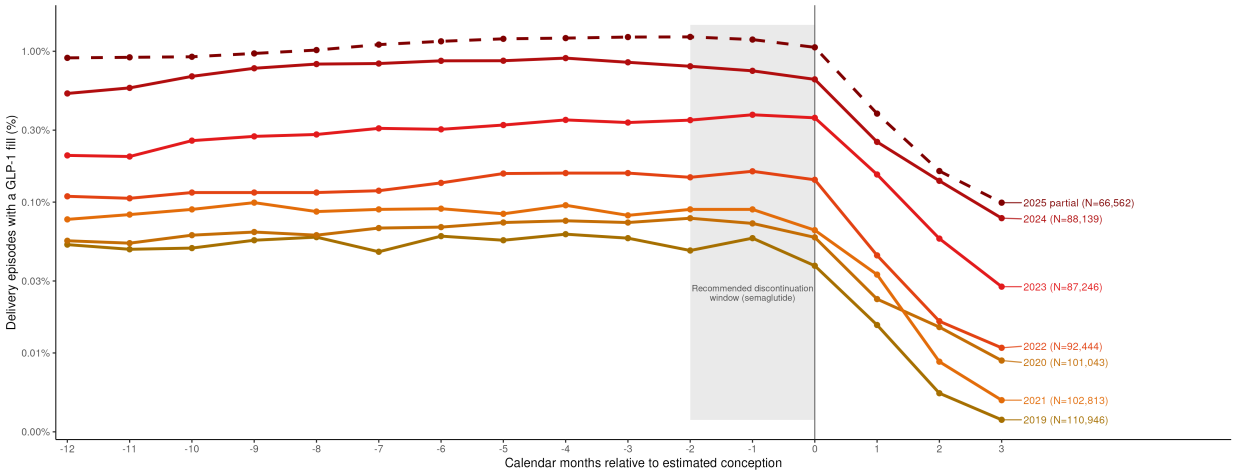



**Figure legend.** Percentage of inpatient live-birth delivery episodes with at least 1 GLP-1 receptor agonist pharmacy fill in each calendar month relative to conception, by delivery year, on a logarithmic scale. Conception was dated using the Z3A gestational-age diagnosis code linked to the delivery admission: delivery admission date minus (gestational-age weeks − 2) × 7 days. Episodes required continuous enrollment with pharmacy benefit in every month of the observation window, from conception month -12 through conception month +3. Month 0 is the month of conception. The shaded band spans the 2-month preconception interval specified in semaglutide labeling. Delivery year 2025 is shown dashed because data covers January through September only. Merative MarketScan Commercial Claims and Encounters, 2017–2025.

Supplemental Table: GLP-1 Receptor Agonist Fill Timing and Characteristics Relative to Gestational-Age-Dated Conception, by Delivery Year, MarketScan Commercial Claims, 2019–2025

| Characteristic | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | All years |
|---|---|---|---|---|---|---|---|---|
| Delivery episodes, No. | 110,946 | 101,043 | 102,813 | 92,444 | 87,246 | 88,139 | 66,562 | 649,193 |
| Any GLP-1 RA fill in 12 mo before conception, No. (%) | 233 (0.21) | 258 (0.26) | 315 (0.31) | 456 (0.49) | 864 (0.99) | 1,993 (2.26) | 1,948 (2.93) | 6,067 (0.93) |
| *Timing of last preconception fill, %* | | | | | | | | |
| 1-30 d before conception | 26.61 | 27.52 | 27.30 | 33.55 | 39.47 | 32.36 | 42.09 | 35.90 |
| 31-60 d before conception | 11.59 | 16.67 | 12.38 | 15.57 | 15.05 | 14.25 | 13.30 | 14.06 |
| 61-90 d before conception | 9.87 | 10.47 | 13.65 | 10.96 | 9.14 | 9.83 | 9.03 | 9.79 |
| 91-180 d before conception | 23.18 | 21.71 | 21.59 | 19.74 | 18.75 | 21.32 | 16.07 | 19.25 |
| 181-365 d before conception | 28.76 | 23.64 | 25.08 | 20.18 | 17.59 | 22.23 | 19.51 | 21.00 |
| Discontinued >=60 d before conception, % | 62.23 | 56.20 | 60.63 | 51.10 | 45.72 | 53.74 | 44.97 | 50.37 |
| Semaglutide last fill >=60 d before conception, % | 50.00 | 42.00 | 56.76 | 44.76 | 44.90 | 55.54 | 51.26 | 51.47 |
| Any fill in mo 0-2 after conception, % | 21.46 | 25.19 | 24.76 | 30.04 | 37.73 | 31.91 | 38.96 | 33.81 |
| Any fill in month +3 after conception, % | 1.29 | 3.10 | 1.27 | 1.97 | 1.97 | 3.16 | 3.23 | 2.75 |
| *Index molecule at last preconception fill, %* | | | | | | | | |
| Semaglutide (any product) | 5.15 | 19.38 | 35.24 | 54.39 | 64.70 | 67.55 | 57.19 | 56.70 |
| Tirzepatide (any product) | 0.00 | 0.00 | 0.00 | 0.00 | 9.03 | 20.84 | 39.32 | 20.76 |
| Liraglutide, dulaglutide, exenatide, lixisenatide | 94.85 | 80.62 | 64.76 | 45.61 | 26.27 | 11.60 | 3.49 | 22.54 |
| *Preconception diagnosis group, %* | | | | | | | | |
| Type 2 diabetes | 33.91 | 37.98 | 35.24 | 31.80 | 22.69 | 16.66 | 16.22 | 21.05 |
| Prior gestational-diabetes evidence | 0.00 | 3.10 | 0.95 | 1.32 | 1.62 | 1.96 | 1.90 | 1.76 |
| Obesity or overweight, no T2D or prior GDM | 48.50 | 44.57 | 44.76 | 50.00 | 56.25 | 62.57 | 63.40 | 58.76 |
| None documented | 17.60 | 14.34 | 19.05 | 16.89 | 19.44 | 18.82 | 18.48 | 18.43 |

*Notes:* Values are counts, counts (percentages), percentages, or rate ratios, as indicated. The All years column includes the partial 2025 delivery cohort. Conception is delivery admission date minus [(maximum valid Z3A week - 2) x 7 days]. The cohort includes episodes with a valid Z3A week and continuous pharmacy-benefit enrollment from GA-dated month -12 through +3. *Semaglutide last fill >=60 d before conception, %:* Further-supplement row; denominator is episodes with an unambiguous semaglutide last preconception fill. *Any fill in mo 0-2 after conception, %:* Calendar event months 0, +1, and +2; month 0 includes days before and after the exact estimated conception date, so this row is not mutually exclusive with the 1-30-day preconception row. *Any fill in month +3 after conception, %:* Measured only in month +3, the end of the required observation window. *Index molecule at last preconception fill, %:* Episodes with multiple molecules on the last preconception fill date are excluded from the three named molecule rows and reported in the extraction diagnostic. Same-day appearances may include reversal/adjustment claims.

## Supplemental Figure. Robustness of Preconception GLP-1 Utilization Pattern Across Conception-Dating Approaches

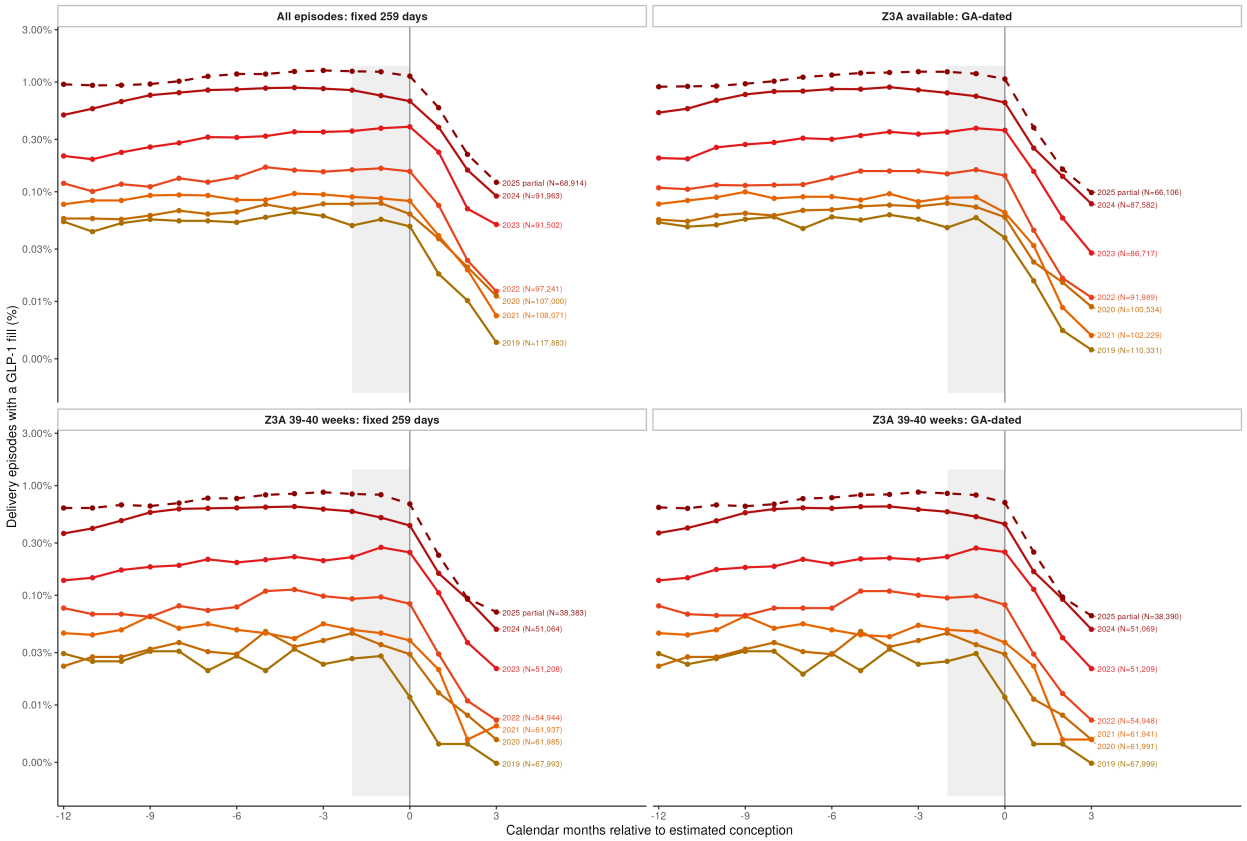



**Figure legend.** Percentage of delivery episodes with a GLP-1 receptor agonist fill in each calendar month relative to conception, by delivery year, on a logarithmic scale, comparing four conception-dating and cohort definitions: all episodes dated by the fixed 259-day method; all episodes with valid Z3A gestational-age evidence, dated using the maximum valid Z3A gestational-age week linked to the delivery admission; episodes whose maximum valid Z3A week was 39–40, dated by the fixed 259-day method; and the same 39–40-week episodes dated using their maximum valid Z3A week. The shaded band spans conception months -2 through 0. Delivery year 2025 is shown dashed because data are partial. The pattern of rising utilization through the preconception year followed by a sharp decline after conception is consistent across all four conception-dating approaches. Merative MarketScan Commercial Claims and Encounters, 2017–2025.

COVER LETTER***** Date Catherine Bradley, MD, MSCE, and Roberto Romero, MD, DMedSci Editors-in-Chief, American Journal of Obstetrics & Gynecology Dear Editors, We submit "Timing of GLP-1 Receptor Agonist Fills Relative to Conception Among Commercially Insured Women" for consideration as a Research Letter. GLP-1 receptor agonist use among reproductive-age women has risen sharply, and semaglutide labeling instructs discontinuation at least 2 months before a planned pregnancy. Whether that instruction is reflected in practice has not been described. Using Merative MarketScan Commercial Claims and Encounters data for 2015 through 2025, we date fills against estimated conception for 649,193 inpatient live-birth delivery episodes and characterize the month-by-month trajectory of fills from 12 months before conception through 3 months after. This analysis is distinguished by dating exposure to conception rather than to delivery, which makes the labeled discontinuation window directly observable. Most of the decline in GLP-1 RA utilization follows conception rather than precedes it, placing cessation at pregnancy recognition rather than at conception planning. The pattern is consistent with a substantial share of these pregnancies being unplanned and points to pregnancy-intention assessment and contraceptive counseling at treatment initiation and dose escalation. The work speaks directly to a conversation AJOG has seen in recent publications. Imbroane et al. (Am J Obstet Gynecol 2025;233:116.e1-116.e7) reported that preconception GLP-1 RA exposure was associated with lower risks of adverse obstetrical outcomes, and the subsequent correspondence in the Journal related to how preconception exposure should be measured. Our contribution is the measurement itself: when, relative to conception, these medications are actually stopped. Space limits kept us to 5 references, so Imbroane is noted here rather than in the reference list. The submission comprises a title page, a body of x words in the four prescribed sections, 5 references, 1 figure, and online supplementary materials (Supplemental Methods, 1 Supplemental Table, and 1 Supplemental Figure). The supplementary materials include 3 additional citations that support methods detail; if the Journal counts these against the 5-reference limit, we will restructure accordingly. The analysis uses deidentified licensed claims data and was determined exempt by the Indiana University Institutional Review Board. The manuscript is not under consideration elsewhere, and all authors have approved the submission. (note if any conflicts, and the funding) Thank you for your consideration. Sincerely, xx

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